

ScikitLearn

LAB(6)

Part-2

Case Study 1: Missing Value Imputation (Iris Dataset)

- Objective: Handle missing values using imputation strategies
- Dataset: `load_iris()` with artificially introduced NaNs
- Tasks:
 - Load Iris dataset and introduce NaNs in every 10th row
 - Compare `SimpleImputer` strategies ('mean', 'median', 'most_frequent')
 - Calculate percentage of values changed during imputation

Questions:

- When should you use median vs. mean imputation?
- Why is `fit_transform()` used instead of `fit() + transform()`?
- Extension: Implement KNN imputation using `KNNImputer`

Case Study 2: Feature Scaling (Wine Dataset)

- Objective: Compare scaling techniques and their impact
- Dataset: `load_wine()`
- Tasks:
 - Apply `StandardScaler`, `MinMaxScaler`, and `RobustScaler`
 - Create histograms comparing original vs. scaled features
 - Calculate and compare standard deviations before/after scaling

Questions:

- Why does `StandardScaler` not guarantee [0,1] range?
- When would `RobustScaler` outperform others?

- Extension: Build pipeline combining scaling and PCA

Case Study 3: Classification Metrics (Breast Cancer Dataset)

- Objective: Evaluate classifier using multiple metrics
- Dataset: `load_breast_cancer()`
- Tasks:
 - Train-test split (80-20)
 - Train Logistic Regression and Random Forest classifiers
 - Generate confusion matrix and calculate precision/recall/F1
 - Plot ROC curves and compute AUC scores

Questions:

- Why might recall be more important than precision for cancer diagnosis?
- What does $AUC=0.5$ indicate?
- Extension: Implement stratified K-fold cross-validation

Case Study 4: Feature Selection (Diabetes Dataset)

- Objective: Select features using filter and wrapper methods
- Dataset: `load_diabetes()`
- Tasks:
 - Use `SelectKBest` with mutual information and f-regression
 - Apply RFE (Recursive Feature Elimination) with Linear Regression
 - Compare selected features from both methods
 - Train models with original vs. selected features

Questions:

- Why does RFE take longer than SelectKBest?
- How does feature selection impact model interpretability?
- Extension: Use permutation importance for feature selection

Case Study 5: Outlier Handling (Boston Housing)

- Objective: Detect and handle outliers
- Dataset: Boston Housing from OpenML
- Tasks:
 - Visualize feature distributions with boxplots
 - Detect outliers using Isolation Forest and Z-score
 - Compare model performance before/after outlier removal

Questions:

- Why is 3.0 a common Z-score threshold?
- How does contamination parameter affect Isolation Forest?
- Extension: Implement DBSCAN for outlier detection

Case Study 6: External Dataset (Heart Disease)

- Objective: Handle real-world dataset challenges
- Dataset: UCI Heart Disease (CSV from URL)
- Tasks:
 - Handle missing values and incorrect entries
 - Encode categorical variables (OneHot vs Label Encoding)
 - Create age groups using binning
 - Generate correlation heatmap

Questions:

- Why convert '?' to NaNs instead of direct imputation?
- When should you avoid one-hot encoding?
- Extension: Implement feature engineering (e.g., BMI calculation)

Case Study 7: Regression Metrics (Housing Prices)

- Objective: Evaluate regression models comprehensively
- Dataset: California Housing dataset
- Tasks:
 - Clean data (handle missing values, remove duplicates)
 - Train Linear Regression and Decision Tree models
 - Calculate MAE, MSE, RMSE, R^2 , and MAPE
 - Visualize residuals

Questions:

- Why is RMSE more interpretable than MSE?
- What does negative R^2 indicate?
- Extension: Implement quantile regression

Case Study 8: Recursive Feature Elimination (Digits)

- Objective: Optimize feature set with wrapper methods
- Dataset: `load_digits()`
- Tasks:
 - Apply RFE with SVM and Random Forest estimators
 - Track accuracy vs number of features
 - Visualize important pixels as heatmap

Questions:

- Why use step=1 in RFE for image data?
- How does estimator choice affect feature selection?
- Extension: Compare with SelectFromModel

Case Study 9: Processing Pipeline (Iris)

- Objective: Build end-to-end preprocessing pipeline
- Dataset: load_iris() with synthetic noise
- Tasks:
 - Create pipeline with:
 - Imputation → Scaling → PCA
 - Imputation → Polynomial Features → Feature Selection
 - Compare pipeline performance vs manual steps
 - Visualize PCA-transformed data

Questions:

- Why should you fit imputer on training data only?
- How does pipeline prevent data leakage?
- Extension: Add custom transformer for outlier removal

Case Study 10: Categorical Encoding (Titanic)

- Objective: Handle high-cardinality categorical features
- Dataset: Titanic survival dataset
- Tasks:
 - Compare:
 - One-Hot Encoding
 - Target Encoding

- Binary Encoding
- Handle rare categories (frequency < 1%)
- Measure impact on Logistic Regression performance

Questions:

- Why might target encoding cause overfitting?
- When should you use hashing vs. binning?
- Extension: Implement CatBoost encoding